

Idea A made precise: line passes on the fine field of one axis

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This replaces the paragraph “Idea A” of `notes/stack-notes.tex`. It refers to the pinned upstream sections `02-streams.tex` (machine model, Lemma elementary-streams and its counter paragraph) and `07-resampling.tex` (lem:tensor-resampling), to `notes/segmented-inverse.tex` and to PR #5’s chirped-correlation blocks. A machine-level simulation of every data movement below is `scripts/check_fine_field.py`; it checks outputs against the linewise maps and checks the movement bounds of Lemma 1 case by case.

1 Setting

Fix an axis i of the padded box. Its box length is $n = t_i \in \{r/2, r\}$, a power of two. Collapse the address fields before axis i to one field X and those after it to one field Y (upstream’s collapsing convention). An entry is a record of $b = O(p)$ bits. Movement is counted in records; a record move costs b head steps. Put $V = |X|n|Y|$ (records), choose f with $2^f \mid n$, write the axis coordinate as

$$j = J2^f + u, \quad 0 \leq u < 2^f, \quad 0 \leq J < n_s := n/2^f,$$

and call (X, J, Y, u) , lexicographic with X most significant, the *slab order*. For fixed X and J the $|Y|2^f$ records with that (X, J) form the *slab* (X, J) ; for fixed Y in it, the 2^f consecutive records form the *slab row* (X, J, Y) . Put $B = |Y|2^f$ and $G = n_s B$ (records per X -group).

A line (X, Y) is *valid* when every coordinate held in X and Y lies below its current bound (s_h or t_h , per the shape descriptor). Invalid lines are zero and must be mapped to zero lines (upstream proof of lem:tensor-resampling).

Definition 1 (Slab-local sweep). A *forward sweep* on a line of box length n is given by

- an input period $\pi \in \{s_i, t_i\}$: the input x is read only through the virtual line $\tilde{x}_j = x_{j \bmod \pi}$, $j \in \mathbb{Z}$, so box entries $j \in [\pi, n)$ are never read;
- an output count $m \leq n$: outputs $k \in [0, m)$ are computed and $k \in [m, n)$ are written 0;
- a non-decreasing centre $c : \mathbb{Z} \rightarrow \mathbb{Z}$ with $c(0) = 0$ and $c(k+a) - c(k) \leq \sigma a + 1$ for $a \geq 0$, where $\sigma \leq 5/4$, and a half-width W such that output k reads $\tilde{x}$ only on $[c(k) - W, c(k) + W]$;
- a state of Σ records per line, an auxiliary row of ξ records per slab row, and per-slab data D_J that depend on J and the axis only (not on X, Y);
- a kernel K_J that, given $\tilde{x}$ on the window of Claim 1, the state after slab $J - 1$ and D_J , returns the 2^f outputs of slab J , the state after slab J and the auxiliary row of slab J .

A *backward sweep* is the same with J decreasing, its state seeded from the final forward state, and its kernel reading the auxiliary row that the forward sweep wrote for the same slab row. The *linewise map* is the sweep run on one line in natural order.

The passes Idea A must carry are instances (Section 3.7): S' and A ($\pi = s_i$, $m = t_i$, $c(k) = \lfloor s_i k / t_i \rfloor$), B_0 ($\pi = t_i$, $m = s_i$, $c(k) = \lfloor t_i k / s_i \rfloor$, so $\sigma = t_i / s_i \leq 5/4$), N , M^{-1} and E ($\pi = s_i$, $m = s_i$, $c(k) = k$), and the Woodbury forward and backward sweeps of the segmented inverse.

2 The lemma

Lemma 1 (Fine-field line passes). *Let a forward sweep satisfy*

$$(W1) \ 2^f \geq 4W + 4;$$

$$(W2) \ \pi \geq 3 \cdot 2^f;$$

$$(W3) \ \Sigma \leq 2^f \text{ and } \xi \leq 2^f.$$

On a fixed set of work tapes (independent of d , $|X|$, $|Y|$, n) the following hold, with all counters, rewinds and returns included.

- (a) *Exposure.* $(X, J, u, Y) \rightarrow (X, J, Y, u)$ and its inverse are f digit moves each and cost exactly $6fV$ record moves each.
- (b) *Forward pass.* The schedule of Section 3.4 writes, in slab order, exactly the outputs of the linewise map on every valid line and zeros elsewhere, and moves at most

$$28V + 11|X|(n_e - n_s)B + \frac{4\Sigma + 2\xi + 2}{2f}V + 4|X||Y|\Sigma$$

records, where $n_e \leq n_s + 4$ is the number of extended slabs.

- (c) *Backward pass.* The schedule of Section 3.5 is likewise exact and moves at most $11V + \frac{3\xi + 6\Sigma + 2}{2f}V + 4|X||Y|\Sigma$ records.
- (d) *Setup.* The per-slab data are computed once per axis and read in lockstep with J ; the validity bitmap costs at most $2|Y|$ moves to build per pass. Kernel work is the linewise work plus at most two partial blocks per slab row.

Hence an axis with P_f forward-type and P_b backward-type passes costs

$$bV(12f + 28P_f + 11P_b + o(1)) \text{ head steps}$$

plus the line algorithm's own arithmetic, when $n_s \rightarrow \infty$; for every axis this is $O(Vb(f + P))$ with $P = P_f + P_b = O(1)$.

The constants are worst-case bounds; the simulation's measured values are listed at the end.

3 Proof

3.1 Exposure and restoration

Write $u = (u_{f-1} \dots u_0)$ in binary. Before step $\iota = 0, \dots, f-1$ the order is $(X, J, u_{f-1} \dots u_\iota, Y, u_{\iota-1} \dots u_0)$. Step ι moves digit u_ι past Y : with prefix $P = |X|n_s 2^{f-1-\iota}$ and suffix $S = 2^\iota$ it maps $(P, u_\iota, Y, S) \rightarrow (P, Y, u_\iota, S)$, which is item 2 of Lemma elementary-streams. Implemented as split on u_ι into two stream tapes (V reads, V writes), return of both streams (V in total), merge in output order (V reads, V writes) and return of the output (V), it costs $6V$. Restoration runs the inverse moves for $\iota = f-1, \dots, 0$. When Y is empty (the last axis) nothing moves.

3.2 The window

Claim 1. Put $J'(J) = \lfloor (c(J2^f) - W)/2^f \rfloor$. Under (W1), every input read by an output of slab J lies in virtual slabs $J', J' + 1, J' + 2$, and $0 \leq J'(J + 1) - J'(J) \leq 2$.

Proof. The inputs of slab J lie in $[c(J2^f) - W, c(J2^f + 2^f - 1) + W]$, an interval of length at most $\sigma(2^f - 1) + 2W + 2 \leq \frac{5}{4}2^f + \frac{1}{2}2^f < 2 \cdot 2^f + 1$ by (W1). It starts at offset $o \in [0, 2^f)$ from $J'2^f$, so it ends before $J'2^f + o + 2 \cdot 2^f + 1 \leq (J' + 3)2^f$. Monotonicity of c gives $J'(J + 1) \geq J'(J)$, and $c((J + 1)2^f) - c(J2^f) \leq \sigma 2^f + 1 < 2 \cdot 2^f$ gives the step bound. $\square$

Remark (The notes' window was wrong). `stack-notes.tex` centres the window at $J' = \lfloor J/\sigma \rfloor$ and reads slabs $J' - 1, J', J' + 1$. The inputs of slab J end near $(J + 1)2^f/\sigma + W$, which reaches slab $J' + 2$ whenever $\{J/\sigma\} > 1 - 1/\sigma - W2^{-f}$, a positive fraction of J since θ is small. For $(s, t, 2^f, W) = (1009, 1024, 64, 8)$, 6 of 16 slabs fail (`scripts/check_fine_field.py`). Anchoring at the lowest input, as above, keeps three slabs.

3.3 Virtual slabs and the wrap splice

For an X-group, the *extended array* holds virtual slabs $v = v_{\text{lo}}, \dots, v_{\text{hi}}$ in order (v, Y, u) , where entry (v, Y, u) is $\tilde{x}_{v2^f+u}$ of line (X, Y) . Here $v_{\text{lo}} = J'(0) = \lfloor -W/2^f \rfloor = -1$ and $v_{\text{hi}} = \max\{J'(J) + 2 : J2^f < m\}$. In every instance $c(m - 1) < \pi$, so $J' \leq n_\pi$, $v_{\text{hi}} \leq n_\pi + 2 \leq n_s + 2$ and $n_e \leq n_s + 4$; and $(v_{\text{hi}} + 1)2^f \leq \pi + 3 \cdot 2^f \leq 2\pi$ by (W2).

Claim 2. Fix v and $\rho = \pi \bmod 2^f$, $n_\pi = \lfloor \pi/2^f \rfloor$. Row (v, Y) is drawn from at most two physical slab rows (a_v, Y) , (b_v, Y) , each as one run of increasing u :

- $0 \leq v < n_\pi$: the row is physical row (v, Y) (aligned copy);
- $v = n_\pi$, $\rho \neq 0$: $u < \rho$ from row (n_π, Y) at u ; $u \geq \rho$ from row $(0, Y)$ at $u - \rho$ (the splice at $u = \pi \bmod 2^f$);
- $v < 0$ or $v > n_\pi$: the virtual range maps to one physical interval of length 2^f starting at $v2^f + u \bmod \pi$, which meets two consecutive slabs, split at the shift ρ .

When $\rho = 0$ (the input period of B_0 is t_i) every virtual row is an aligned copy.

Proof. $[v2^f, (v+1)2^f)$ has length $2^f < \pi$ and lies in $[-\pi, 2\pi)$, so it contains at most one multiple of π . If none, its image mod π is one interval of length 2^f , which meets at most two slabs, each in an increasing run. If it contains π it is $v = n_\pi$, its part below π is $[n_\pi 2^f, \pi)$ inside slab n_π , and its part above maps to $[0, 2^f - \rho)$ inside slab 0. It cannot contain 0 unless $v = 0$, where the range starts at 0. $\square$

Building it. Tape IN holds the array; first copy the X-group to tape CP (G reads, G writes, one return). For each v park IN at row $(a_v, 0)$ and CP at row $(b_v, 0)$. For $Y = 0, \dots, |Y| - 1$ read the two runs in u order, each head finishing its slab row, and write every value to E0, E1, E2 in the same step. Each head advances exactly 2^f per Y , so B per virtual slab. A single head would pay $|Y|2^f$ per row to jump between the two physical slabs, i.e. $\Theta(|Y|B)$ per spliced slab; the second head is what makes the splice linear. The sequence of a_v is $n_\pi - 1$ (for $v = -1$), $0, 1, \dots, n_\pi$, then small indices again, so IN seeks across the group at most four times and CP at most twice.

3.4 Forward schedule

Tapes: IN, CP, E0, E1, E2, OUT, ST (state), VB (validity bitmap over Y), AX (auxiliary rows), FIN (final states), BUF (window buffer), DJ (per-slab data), descriptor and counter tapes.

Validity. VB holds $\prod_h [y_h < \text{bound}_h]$ over Y in Y order. Build it from the innermost Y axis outward: the bitmap for axes $h..$ is the concatenation over digit values of copies of the bitmap for axes $h+1..$, or of zeros above the bound. Each level costs twice its size and sizes at least double, so building costs at most $2|Y|$ per tape. X -validity is computed once per group by an odometer, $O(L)$ against a group of $G \geq n$ records.

For each X in order:

1. If X is invalid, advance IN by G and write $n_s B$ zeros to OUT, $n_s |Y| \xi$ to AX and $|Y| \Sigma$ to FIN.
2. Otherwise build the extended array and write $|Y| \Sigma$ zero states to ST.
3. For $J = 0, \dots, n_s - 1$: if $J2^f \geq m$ write a zero slab and zero auxiliary rows. Otherwise seek Er to row $(J'(J) + r, 0)$ for $r = 0, 1, 2$, and ST, VB to their starts. For each Y : read the VB bit; copy 2^f records from each of E0, E1, E2 into BUF, which then holds $\tilde{x}$ on $[J'2^f, (J' + 3)2^f)$; read the Σ -record state. If valid, run K_J with D_J , step ST back Σ and

write the new state. Write the 2^f outputs (zeros if invalid or $k \geq m$) to OUT and the ξ -record auxiliary row to AX.

4. Copy ST to FIN.

Invariant. At the start of (J, Y) , head Er is at row $(J'(J) + r, Y)$, ST at record Y , VB at bit Y , and OUT, AX at position (X, J, Y) of slab order. It holds at $Y = 0$ by the seeks, and each step advances Er by one row, ST by Σ , VB by one, OUT by 2^f and AX by ξ .

Correctness. By induction on J , state record Y of group X equals the linewise state of line (X, Y) after slabs $0..J - 1$: only step (J, Y) touches record Y , and the slabs of each line are visited in increasing J . By Claims 1 and 2, BUF contains the exact values the linewise kernel reads, so outputs, states and auxiliary rows agree. Invalid lines keep a zero state and get zero outputs, as the linewise rule writes. No padded box entry $j \in [\pi, n)$ is read (Claim 2), and every output position is written once, so the box keeps its full size and slab order.

Movement per valid group. IN: $\leq 6G + n_e B$ (entering the group, copy, return, four cross-group seeks, the build reads). CP: $\leq 3G + n_e B$. Each Er : $n_e B$ writes, a return after the build and one at the next group, and in the main loop B per slab plus a re-seek of at most B (head ends at $J' + r + 1$, needs $J'(J + 1) + r \in [J' + r, J' + r + 2]$), so $\leq 3n_e B + 2n_s B$. OUT: G , plus one final return of V in total. ST: $4|Y|\Sigma$ per slab and $4|Y|\Sigma$ at the ends. VB: $2|Y|$ per slab. AX: $|Y|\xi$ per slab plus a final return. Summing with $G = n_s B$:

$$|X|B(16n_s + 11n_e) + V + \frac{4\Sigma + 2\xi + 2}{2^f}V + 4|X||Y|\Sigma,$$

which is the bound in (b). Invalid groups cost less.

3.5 Backward schedule

Input is the forward output Z (slab order), AX and FIN. For each valid X : copy its $|Y|\Sigma$ final states from FIN to ST; for J from $\lfloor (m - 1)/2^f \rfloor$ down to 0, seek Z and the output tape to slab (X, J) and AX to row $(X, J, 0)$; for each Y read the slab row into BUF and its auxiliary row, read the state, run the backward kernel over u *decreasing* inside BUF, write the new state, and write the 2^f outputs in increasing u . Lines are independent, so visiting Y in increasing order inside a decreasing- J sweep is legitimate; only the order along each line matters. Invariant and induction are as in the forward case with J descending, and AX row (X, J, Y) is read at the position where it was written.

Movement: Z and the output each read or write B and seek back $2B$ per slab, plus entering the group ($\leq 2G$): $\leq 5G$ each, and one final return of the output (V). AX: $3|Y|\xi$ per slab. ST: $\leq 4|Y|\Sigma$ per slab plus loading. VB: $2|Y|$ per slab. This gives (c). If a backward-type kernel needs neighbouring slabs (the second M^{-1} after the backward solve), use the three-tape window of Section 3.4 with J descending: J' is non-increasing with steps at most 2, and the forward bound applies.

3.6 Per-slab data and setup

D_J consists of $J'(J)$, the centres $c(k)$ and fractional offsets for k in slab J , the block boundaries and q -counters, the list of wraps in slab J , and (for Woodbury) the elimination factors of the wraps in slab J . None depends on X or Y . They are produced once per axis by incrementing the q -counters along J , $O(2^f p)$ bit operations per slab, and stored on DJ in J order; DJ is swept once per X-group. Since it is read once per group and not per line, the total is $|X| \cdot |D|$, with $|D|$ the size of all D_J (bounded in Section 3.7). This replaces the notes' per-chunk odometer test, which is no longer needed: validity is one VB bit per slab row. A block of the line algorithm that straddles a slab boundary is recomputed from BUF by both slabs; with blocks of length at most $W \leq 2^f/4$, at most two partial blocks occur per slab row, so kernel work is at most $(1 + 2W/2^f) \leq 3/2$ times linewise.

3.7 Instances and the conditions (W1)–(W3)

- S' , A , B_0 , E : windowed, no state ($\Sigma = \xi = 0$). $W = 3m + 3$ for S' (notes), the halo of E .
- M^{-1} via PR #5's chirped correlation: each segment block has length at most $\lambda = \lfloor 1/\theta \rfloor + 1$ and reads a halo w ; take $W \geq \lambda + 2w$ so every segment meeting slab J lies in BUF and is recomputed there. No state. The segment crossing the period boundary ("stitched from the two line ends" in `segmented-inverse.tex`) is read through $\tilde{x}$, i.e. through the halo slabs $v = -1$ and $v \geq n_\pi$; no separate stitching.
- Woodbury. N is the same matrix on every line of axis i , so $C = I + V^\top M^{-1}U$ and all its elimination factors are line-independent and live in D_J : $O(w^2)$ words of precision P per wrap, $|D| = O((\theta n + 1)w^2 P/b)$ records per axis, read once per group, a fraction $O(\theta w^2 P/b)$ of V even when $|Y| = 1$. Per line only right-hand sides are carried. Forward sweep: the state is the running reduced right-hand side of the

current block row ($2w$ words) and the cyclic-border accumulator for the last block row ($2w$ words), so $\Sigma = 4wP/b$; the auxiliary row stores the reduced right-hand side of each wrap in the slab, $\xi = (\lceil \theta 2^f \rceil + 1)2wP/b$. After slab $n_s - 1$ every line of the group solves its last $2w \times 2w$ border system (factor in D) for x_{last} , kept in FIN. Backward sweep: state $(x_{k+1}, x_{\text{last}})$, $4w$ words; it reads the auxiliary rows in reverse.

- Conditions. With the recentred sub-blocks $P \leq 2Q + O(\log s) = O(b)$. Then (W3) needs $2^f \geq Cw$ and $\theta w \leq c$; with $\alpha^2 \approx Q/(48d)$, $w = O(\sqrt{d})$ and $\theta \approx 1/(4dL^y)$, so $\theta w \rightarrow 0$, and (W1) with $W \geq \lambda + 2w = \Theta(dL^y)$ gives $2^f = \Theta(dL^y)$, $f = O(\log L)$. (W2) holds since $s_i \geq \frac{4}{5}t_i \geq r/4 \gg 2^f$.

3.8 Total

Per axis: exposure and restoration $12fVb$; each forward-type pass at most $28Vb$ and each backward-type pass at most $11Vb$ up to the terms $11|X|(n_e - n_s)B = O(V2^f/n)$, $O(V(\Sigma + \xi)/2^f)$ and $O(|X||Y|\Sigma)$, all $O(V)$ by (W3) and $o(V)$ as $n_s \rightarrow \infty$; the setup terms above are $o(Vb)$. Over d axes this is $O(Vbd(\log L + P))$, as the notes claim. $\square$

4 Changes to the notes' paragraph

1. The centred window $J' - 1 \dots J' + 1$ with $J' = \lfloor J/\sigma \rfloor$ misses inputs; anchor J' at the lowest input read and use $J', J' + 1, J' + 2$ (Claim 1), with $2^f \geq 4W + 4$. J' moves by 0, 1 or 2 per slab (2 occurs for B_0).
2. The wrap splice needs two heads advancing in lockstep over Y (Claim 2); copying “the boundary slabs” with one head costs $\Theta(|Y|)$ per record. The tail splice pairs slab n_π with slab 0, and the halo slabs pair consecutive slabs shifted by $\pi \bmod 2^f$. (W2) $\pi \geq 3 \cdot 2^f$ keeps every virtual slab on at most two physical slabs.
3. Per-line state for all $|Y|$ interleaved lines is an array in Y order on its own tape, swept once per slab; auxiliary data for the backward sweep is an array in (X, J, Y) order. Both must fit in a slab row, which is (W3), and this is what lower-bounds 2^f besides the window.
4. The Woodbury factors are line-independent; only right-hand sides are per line. Storing factors per line (as “per-line state” suggests) would

cost $\Theta(w^2)$ per wrap per line and violate (W3) when $\theta w^2 P/b$ is not small.

5. Per-chunk validity by an odometer over Y is replaced by a precomputed bitmap read in lockstep; the hypothesis “chunk volume $\gg \ell^2$ ” is not needed.

5 Machine check

`scripts/check_fine_field.py` represents every array as a tape with one head and counts every head step. It runs exposure, a forward pass (windowed periodic map with k -dependent weights, a two-word recurrence state, sparse wrap records to AX), a backward pass seeded from FIN that reads AX in reverse, and restoration, for twelve shapes: $s \in \{13, 29, 53, 61, 103, 107, 127\}$ (prime, so $s \bmod 2^f \neq 0$), $2^f \in \{4, 8, 16\}$, $|Y| \in \{1, \dots, 32\}$ with invalid Y coordinates, $|X| \in \{2, 3, 4\}$ with invalid groups, and expansion, compression and $s \rightarrow s$ maps. All outputs equal the linewise maps; every case satisfies the bounds (a)–(c). Measured forward 12.8–25.8 V , backward 3.8–9.6 V , exposure exactly 6 V per bit; on longer lines ($n = 256$ –1024, $2^f = 16$ –64) forward 18.4–21.3 V and backward 9.0–11.3 V , independent of $|Y| \in \{1, 16, 64\}$.